African Institute for Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 9: Properties of Estimators

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Bias of an estimator

Notation: $g_n(X_1, \cdots, X_n) = g_n(\underline{X})$

Definition

*An estimator $g_n(\underline{X})$ of θ is an **unbiased estimator** if*

$$E\left[g_n(X_1, \cdots, X_n)\right] = \theta, \text{ for all } \theta \in \Theta.$$

$b_n(\theta) = E\left[g_n(\underline{X})\right] - \theta$ is called the **bias** of the estimator $g_n(\underline{X})$.

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Exercise

- Let $X_i \sim \mathscr{B}(N, p)$, $i = 1, \ldots, n$.
  Is $\widehat{p} = \overline{X}_n / N$, the MLE, an unbiased estimator?

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  Is $\widehat{p} = \overline{X}_n/N$, the MLE, an unbiased estimator?
- Let $X_i \sim \mathscr{N}(\mu, \sigma^2)$, $i = 1, \ldots, n$.
  Is $s_X^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$ an unbiased estimator of σ^2?

Bias of an estimator

Definition

*We say that $g_n(\underline{X})$ is an **asymptotically unbiased estimator** of θ if*

$$\lim_{n \to +\infty} b_n(\theta) = 0, \ \textit{for all } \theta \in \Theta.$$

Exercise:

Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$, $i = 1, \ldots, n$.

- Show that s_X^2 is an asymptotically unbiased estimator of σ^2.

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- Show that s_X^2 is an asymptotically unbiased estimator of σ^2.
- Build an unbiased estimator of σ^2.

$$\hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X}_n)^2 \text{ is an unbiased estimator of } \sigma^2.$$

Convergence of a sequence of r.v.'s

Let Z_n be a sequence of r.v. ➥ $g_n(\underline{X})$ is a sequence of r.v.

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Convergence in probability

Z_n converges to Z **in probability** iff,

$$\forall \varepsilon > 0, \lim_{n \to +\infty} P\left(\left|Z_n - Z\right| > \varepsilon\right) = 0.$$

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Notation: $Z_n \overset{P}{\to} Z$

Convergence **in quadratic mean** or in mean square

Z_n converges to Z in quadratic mean $\left(Z_n \overset{q.m.}{\to} Z\right)$ iff,

$$\forall \varepsilon > 0, \lim_{n \to +\infty} E\left(Z_n - Z\right)^2 = 0.$$

Notation: $Z_n \overset{q.m.}{\to} Z$

Convergence in Probability

Definition

Let $X_1, \cdots, X_n$ be n r.v.'s with a distribution P_θ, $\theta \in \Theta$, where θ is unknown belonging to Θ, then the sequence $g_1, g_2, \ldots$ is called a ***consistent*** *sequence of estimators of θ if*

$$g_n(\underline{X}) \text{ converges in probability to } \theta \quad \left(g_n(\underline{X}) \xrightarrow{P} \theta\right)$$

that is:

$$\forall \varepsilon > 0, \ \lim_{n \to +\infty} P\left(\left|g_n(\underline{X}) - \theta\right| > \varepsilon\right) = 0, \ \forall \theta \in \Theta.$$

$g_n(\underline{X})$ is said to be a ***consistent estimator*** *of θ.*

Convergence in Probability

Example:

Let $X_1, \cdots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\overline{X}_n$ converge in probability to μ.

That is to say:

$$\forall \varepsilon > 0, \quad \lim_{n \to +\infty} P\left(\left|\overline{X}_n - \mu\right| > \varepsilon\right) = 0.$$

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That is to say:

$$\forall \varepsilon > 0, \quad \lim_{n \to +\infty} P\left(\left|\overline{X}_n - \mu\right| > \varepsilon\right) = 0.$$

Proof: $\overline{X}_n \sim \mathcal{N}\left(\mu, \sigma^2/n\right)$ and $\dfrac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} = Z_n \sim \mathcal{N}\left(0, 1\right)$.

Then

$$\begin{aligned} P\left(\left|\overline{X}_n - \mu\right| > \varepsilon\right) &= P\left(\left|Z_n\right| > \frac{\varepsilon}{\sigma}\sqrt{n}\right) = 1 - P\left(\left|Z_n\right| \leqslant \frac{\varepsilon}{\sigma}\sqrt{n}\right) \\ &= 2 - 2\Phi\left(\frac{\varepsilon}{\sigma}\sqrt{n}\right). \end{aligned}$$

But $\lim_{n \to +\infty} \Phi\left(\frac{\varepsilon}{\sigma}\sqrt{n}\right) = 1$; that shows the result.

Two useful inequalities:

Markov's Inequality

Let X be a positive r.v. such that $EX < +\infty$. Then:

$$\forall \delta > 0, \; P(X > \delta) \leqslant \frac{EX}{\delta}.$$

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Chebyshev's Inequality

Let X be a r.v. such that $EX < +\infty$ and $Var(X) < +\infty$. Then:

$$\forall \varepsilon > 0,\ P(|X - EX| > \varepsilon) \leqslant \frac{Var(X)}{\varepsilon^2}.$$

Proof: Apply the Markov's Inequality to $Y = (X - EX)^2$.

Convergence in Probability

Exemple (cont.):

Let $X_1, \cdots, X_n$, be n independent r.v. such that $X_i \sim \mathscr{N}(\mu, \sigma^2)$, then $\overline{X}_n$ converge in probability to μ.

Convergence in Probability

Exemple (cont.):

Let $X_1, \cdots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\overline{X}_n$ converge in probability to μ.

Applying Chebytshev's inequality to $\overline{X}_n$:

$$P(|\overline{X}_n - \mu| > \varepsilon) \leqslant \frac{\sigma^2}{n\varepsilon} \longrightarrow 0. \qquad \Rightarrow \quad \overline{X}_n \xrightarrow{P} \mu.$$

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Exercise

Let $X_1, \cdots, X_n$, be n independent r.v. such that $X_i \sim \mathscr{P}(\lambda)$. Show that $\overline{X}_n$ converge in probability to λ.

Consistency in quadratic mean

Definition

Let $X_1, \cdots, X_n$ be n r.v.'s with a distribution P_θ, $\theta \in \Theta$.
Let $g_n(\underline{X})$ be a statistic of $\underline{X}$.
We say that $g_n(\underline{X})$ is ***consistent in quadratic mean*** $\left(g_n(\underline{X}) \stackrel{q.m.}{\rightarrow} \theta\right)$ *if*

$$\lim_{n \to +\infty} E\left[(g_n(\underline{X}) - \theta)^2\right] = 0, \ \forall \theta \in \Theta.$$

The quantity $E\left[(g_n(\underline{X}) - \theta)^2\right]$ is called the **mean squared error (MSE)** of the estimator.

Consistency in quadratic mean

Exemple:

Let $X_1, \cdots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\overline{X}_n$ converge in quadratic mean to μ.

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Let $X_1, \cdots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\overline{X}_n$ converge in quadratic mean to μ.

Proof:

$$E(\overline{X}_n - \mu)^2 = Var(\overline{X}_n) = \sigma^2/n \longrightarrow 0. \quad ➡ \quad \overline{X}_n \overset{q.m.}{\to} \mu.$$

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Let $X_1, \cdots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\overline{X}_n$ converge in quadratic mean to μ.

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Exercise

Let $X_1, \cdots, X_n$, be n independent r.v. such that $X_i \sim \mathscr{P}(\lambda)$. Show that $\overline{X}_n$ converge in quadratic mean to λ.

Consistency in quadratic mean

Theorem

$g_n(\underline{X})$ *is* ***consistent in quadratic mean*** *iff*

(i) $\lim_{n \to +\infty} Var\left[g_n(\underline{X})\right] = 0$

(ii) $g_n(\underline{X})$ *is asymptotically unbiased.*

Proof:

Lemma

$$E\left[\left(g_n(\underline{X}) - \theta\right)^2\right] = Var\left[g_n(\underline{X})\right] + \left[b_n(\theta)\right]^2, \quad \forall \theta \in \Theta.$$

➥ *Bias-Variance Trade-Off*

Consistency in quadratic mean

Theorem

*$g_n(\underline{X})$ is consistent **in quadratic mean**, $g_n(\underline{X})$ converges **in probability**.*

$$\left(g_n(\underline{X}) \xrightarrow{q.m.} \theta\right) \Longrightarrow \left(g_n(\underline{X}) \xrightarrow{P} \theta\right).$$

Proof:

Apply Markov's inequality to $|g_n(\underline{X}) - \theta|^2$.